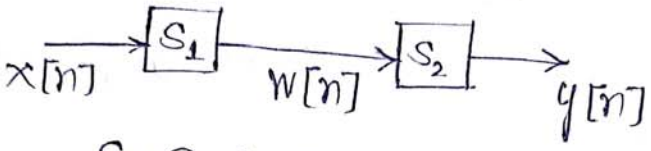
Sol 1(a):

Given that – $w[n] = \frac{1}{2} w[n-1] + x[n]$ —— (1)

$y[n] = \alpha\, y[n-1] + \beta\, w[n]$ —— (2)

$y[n] = -\frac{1}{8} y[n-2] + \frac{3}{4} y[n-1] + x[n]$ —— (3)



S_1, S_2 : Causal LTI system

$\therefore y[n] = \alpha\, y[n-1] + \beta\, w[n]$

From this we may write,

$$w[n] = \frac{1}{\beta} y[n] - \frac{\alpha}{\beta} y[n-1] \quad \text{—— (4)}$$

and $$w[n-1] = \frac{1}{\beta} y[n-1] - \frac{\alpha}{\beta} y[n-2] \quad \text{—— (5)}$$

Multiplying both side of eqn (5) with $(\frac{1}{2})$, we get –

$$\frac{1}{2} w[n-1] = \frac{1}{2\beta} y[n-1] - \frac{\alpha}{2\beta} y[n-2] \quad \text{—— (6)}$$

Subtracting eqn (4) & eqn (6), we get –

$$w[n] - \frac{1}{2} w[n-1] = \frac{1}{\beta} y[n] - \frac{\alpha}{\beta} y[n-1] - \frac{1}{2\beta} y[n-1] + \frac{\alpha}{2\beta} y[n-2] \quad \text{—— (7)}$$

By eqn (1) & eqn (7), we get –

$$x[n] = \frac{1}{\beta} y[n] - \frac{\alpha}{\beta} y[n-1] - \frac{1}{2\beta} y[n-1] + \frac{\alpha}{2\beta} y[n-2]$$

$$y[n] = \left(\alpha + \frac{1}{2}\right) y[n-1] - \left(\frac{\alpha}{2}\right) y[n-2] + \beta\, x[n] \quad \text{—— (8)}$$

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Comparing eqn(3) and eqn(8), we get —

$$\left(\alpha + \frac{1}{2}\right) = \left(\frac{3}{4}\right) \quad \text{and} \quad \beta = 1.$$

$$\alpha = \left(\frac{1}{4}\right) \quad \text{and} \quad \beta = 1. \qquad \rightarrow (2 \times 2 \text{ Marks})$$

Sol 1(b):

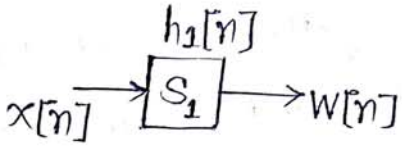
Let impulse response of system S_1 & S_2 are $h_1[n]$ & $h_2[n]$ respectively.

By eqn(1), $W[n] = \frac{1}{2} W[n-1] + x[n]$

$$h_1[n] - \frac{1}{2} h_1[n-1] = \delta[n]$$

$\therefore h_1[n] = \left(\frac{1}{2}\right)^n u[n]$ —— (9)

$\rightarrow$ (1 Marks)



$= \delta[n]$

$W[n] = x[n] * h_1[n]$

$= \delta[n] * h_1[n]$

$= h_1[n]$

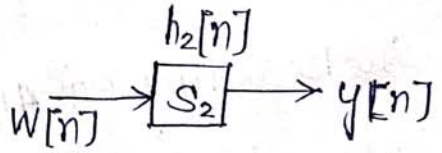
By eqn(2), $y[n] = \alpha y[n-1] + \beta W[n]$

$$y[n] = \frac{1}{4} y[n-1] + W[n]$$

$$h_2[n] - \frac{1}{4} h_2[n-1] = \delta[n]$$

$\therefore h_2[n] = \left(\frac{1}{4}\right)^n u[n]$ —— (10)

$\rightarrow$ (1 Marks)



$= \delta[n]$

$y[n] = W[n] * h_2[n]$

$= \delta[n] * h_2[n]$

$= h_2[n]$

$\therefore$ Impulse response of cascaded systems,

$$h[n] = h_1[n] * h_2[n]$$

$$= \sum_{k=-\infty}^{\infty} h_1[k] \cdot h_2[n-k]$$

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$$\therefore h[n] = \sum_{K=0}^{\infty} \left(\frac{1}{2}\right)^K \left(\frac{1}{4}\right)^{n-K} u[n-K]$$

$$= \sum_{K=0}^{n} \left(\frac{1}{2}\right)^K \left(\frac{1}{4}\right)^{n-K} = \left(\frac{1}{4}\right)^n \sum_{K=0}^{n} \left(\frac{1}{2}\right)^K \cdot \left(\frac{1}{4}\right)^{-K}$$

$$= \left(\frac{1}{4}\right)^n \sum_{K=0}^{n} \left(\frac{1}{2}\right)^{-K} = \left(\frac{1}{4}\right)^n \cdot \frac{1\left[2^{n+1} - 1\right]}{[2-1]}$$

$$= \left(\frac{1}{2}\right)^{2n} \cdot \left[2^{n+1} - 1\right]$$

$$= \left[2\left(\frac{1}{2}\right)^n - \left(\frac{1}{4}\right)^n\right] u[n]$$ → (2 Marks)

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Sol(2):- Given impulse response of an LTI system,

$$h(t) = e^{-6|t|}$$

$$= \begin{cases} e^{-6t}, & t>0 \\ e^{-6(-t)}, & t<0 \end{cases}$$

$$= e^{-6t}\cdot u(t) + e^{6t}\cdot u(-t) \quad \text{—(1)}$$

Step(I):- condition for LTI system to be causal,

$$h(t) = 0 \quad , \quad t<0$$

By eqn(1), $h(t) \neq 0$ for $t<0$, so the given LTI system is Non-causal.

→ (2 Mark)

Step(II):- Condition for LTI system to be stable, impulse response of the system should be absolutely integrable.

i.e. $\int_{-\infty}^{\infty} |h(t)|\,dt < \infty$

By eqn(1), $h(t) = e^{-6t}u(t) + e^{6t}u(-t)$

$$\int_{-\infty}^{\infty} |h(t)|\,dt = \int_{-\infty}^{\infty} |e^{-6t}u(t) + e^{6t}u(-t)|\,dt$$

$$= \int_{-\infty}^{0} e^{6t}dt + \int_{0}^{\infty} e^{-6t}dt = \left(\frac{e^{6t}}{6}\right)_{-\infty}^{0} + \left(\frac{e^{-6t}}{-6}\right)_{0}^{\infty}$$

$$= \left(\frac{1-0}{6}\right) + \left(\frac{0-1}{-6}\right) = \left(\frac{1}{3}\right)$$

So, the given LTI system is stable.

→ (2 Mark)

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Sol 3(I):- Given that- $x(t) = 3\cos\left(4t + \frac{\pi}{3}\right)$ ——— (1)

For Periodic signal,

$$x(t+T) = x(t) \quad \forall t$$

where, T = FTP (smallest positive)

} (0.5 Mark)

$$x(t+T) = 3\cos\left\{4(t+T) + \frac{\pi}{3}\right\}$$

$$= 3\cos\left\{4t + \frac{\pi}{3} + 4T\right\}$$

if $T = \frac{\pi}{2}$

then— $$x(t+T) = 3\cos\left\{4t + \frac{\pi}{3} + 4\times\frac{\pi}{2}\right\}$$

$$= 3\cos\left\{4t + \frac{\pi}{3} + 2\pi\right\}$$

$$= 3\cos\left(4t + \frac{\pi}{3}\right)$$

$$= x(t)$$

→ (1.5 Mark)

Hence, $x(t)$ is periodic signal with FTP, $T = \frac{\pi}{2}$

→ (1 Mark)

Sol 3(II):- Given that- $$x[n] = \cos\left(\frac{\pi}{2}n\right)\cos\left(\frac{\pi}{4}n\right)$$

$$= \frac{1}{2}\left\{\cos\left(\frac{\pi}{2}n + \frac{\pi}{4}n\right) + \cos\left(\frac{\pi}{2}n - \frac{\pi}{4}n\right)\right\}$$

$$= \frac{1}{2}\left\{\cos\left(\frac{3\pi}{4}n\right) + \cos\left(\frac{\pi}{4}n\right)\right\}$$

$$= \frac{1}{2}\cos\left(\frac{3\pi}{4}n\right) + \frac{1}{2}\cos\left(\frac{\pi}{4}n\right)$$

$$= x_1[n] + x_2[n]$$ ——— (2)

For Periodic signal,

$$x[n+N] = x[n] \quad \forall n$$

where, N = FTP (smallest integer)

} (0.5 Mark)

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By eqn(2), $x_1[n] = \frac{1}{2}\cos\left(\frac{3\pi}{4}n\right)$

$$x_1[n+N_1] = \frac{1}{2}\cos\left\{\frac{3\pi}{4}(n+N_1)\right\}$$

$$= \frac{1}{2}\cos\left\{\frac{3\pi}{4}n + \frac{3\pi}{4}N_1\right\}$$

if $N_1 = 8$ then — $x_1[n+N_1] = x_1[n]$

Hence, $x_1[n]$ is periodic signal with FTP, $N_1 = 8$ —— (3)

→ (1 Mark)

By eqn(2), $x_2[n] = \frac{1}{2}\cos\left(\frac{\pi}{4}n\right)$

$$x_2[n+N_2] = \frac{1}{2}\cos\left\{\frac{\pi}{4}(n+N_2)\right\}$$

$$= \frac{1}{2}\cos\left\{\frac{\pi}{4}n + \frac{\pi}{4}N_2\right\}$$

if $N_2 = 8$ then — $x_2[n+N_2] = x_2[n]$

Hence, $x_2[n]$ is periodic signal with FTP, $N_2 = 8$ —— (4)

→ (1 Mark)

By eqn(2), (3) & (4), we get —

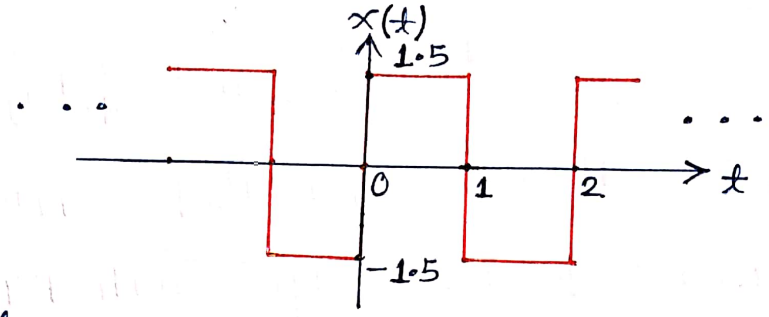
$x[n]$ is periodic signal with FTP, $N = LCM(N_1, N_2) = 8$

→ (0.5 Mark)

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Sol(4):- Given that –

$$x(t) = \begin{cases} 1.5, & 0 \le t < 1 \\ -1.5, & 1 \le t < 2 \end{cases}$$



(a) Fundamental Time period of signal $x(t)$, $T_0 = 2$

Fundamental freq., $\omega_0 = \frac{2\pi}{T_0} = \frac{2\pi}{2} = \pi$ rad/sec → (1 Mark)

(b) As we know – Fourier series coefficients a_k for periodic signal $x(t)$,

$$a_k = \frac{1}{T_0}\int_{T_0} x(t)\cdot e^{-jk\omega_0 t}\cdot dt$$

Case(I):- a_k for $k=0$, $a_0 = \frac{1}{T_0}\int_{T_0} x(t)\cdot e^{0}\cdot dt$

$$= \frac{1}{2}\int_0^2 x(t)\,dt = 0$$ → (1 Mark)

Case(II):- a_k for $k\neq 0$, $a_k = \frac{1}{T_0}\int_{T_0} x(t)\cdot e^{-jk\omega_0 t}\cdot dt$

$$= \frac{1}{2}\int_0^2 x(t)\cdot e^{-jk\cdot\pi\cdot t}\cdot dt$$

$$= \frac{1}{2}\left\{\int_0^1 1.5\, e^{-jk\pi t}\cdot dt + \int_1^2 (-1.5)\, e^{-jk\pi t}\cdot dt\right\}$$

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$$a_k = \frac{1}{2}\left\{ 1.5\left(\frac{e^{-j\pi kt}}{-j\pi k}\right)_0^1 - 1.5\left(\frac{e^{-j\pi kt}}{-j\pi k}\right)_1^2 \right\}$$

$$= \frac{1}{2}\left\{ \left(\frac{1.5}{-j\pi k}\right)\left(e^{-j\pi k} - 1\right) - \left(\frac{1.5}{-j\pi k}\right)\left(e^{-j\pi k\cdot 2} - e^{-j\pi k}\right) \right\}$$

$$= \frac{1}{2}\left(\frac{1.5}{-j\pi k}\right)\left\{ e^{-j\pi k} - 1 - e^{-j2\pi k} + e^{-j\pi k} \right\}$$

$$= \frac{1}{2}\left(\frac{1.5}{-j\pi k}\right)\left\{ 2e^{-j\pi k} - e^{-j2\pi k} - 1 \right\}$$

$$= \frac{1}{2}\left(\frac{1.5}{-j\pi k}\right)\left(2e^{-j\pi k} - 1 - 1\right) = \frac{1}{2}\left(\frac{1.5}{-j\pi k}\right)\left(2e^{-j\pi k} - 2\right)$$

$$= \left(\frac{1.5}{j\pi k}\right)\left(1 - e^{-j\pi k}\right)$$

$$= \left(\frac{3}{k\pi}\right)\left(\frac{e^{-j\frac{\pi k}{2}} \cdot e^{j\frac{\pi k}{2}} - e^{-j\frac{\pi k}{2}} \cdot e^{-j\frac{\pi k}{2}}}{2j}\right)$$

$$= \left(\frac{3}{k\pi}\right) e^{-j\frac{\pi k}{2}} \cdot \sin\left(\frac{\pi k}{2}\right)$$

→ (4 Mark)

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